Notes: Mayer, Melitz, and Ottaviano (AER, 2014)
Market Size, Competition, and the Product Mix of Exporters

Contents

1 Big picture

- **Question.** How do export-market size and competition affect the product range and product mix chosen by multi-product exporters?
- **Core idea.** Multi-product firms do not simply decide whether to export to a destination. They also decide which products to export and how intensively to sell each product. Destination-level competition changes these within-firm product allocations.
- **Mechanism.** Tougher competition lowers markups for all products but hurts worse products more. This causes firms to skew sales toward their best-performing products and to drop marginal products.
- **Theory.** The paper embeds multi-product firms into a Melitz–Ottaviano-style model with variable markups and asymmetric export destinations. A firm's products differ by distance from the firm's core competence.
- **Empirical setting.** The authors use French customs microdata at the firm-product-destination level. The key empirical comparisons are within the same firm across export destinations.
- **Main empirical result.** Export sales are more skewed toward core products in larger destinations and in destinations with greater supply potential, both of which proxy for tougher competition.
- **Product-mix fact.** A given firm sells relatively more of its core product compared with its second-best product in tougher markets.
- **Beyond the top products.** The result holds for measures of skewness across the full set of products exported by a firm to a destination.
- **Productivity implication.** Because firms shift output toward better products in more competitive environments, competition changes measured firm productivity even holding the firm's underlying product technologies fixed.
- **Takeaway.** Trade and competition affect productivity not only through selection across firms, but also through product-mix reallocation within multi-product firms.

2 Incremental contribution of the paper

2.1 Contribution relative to standard heterogeneous-firm trade models

- Standard Melitz-type models focus on firm entry, export participation, and selection across firms.
- This paper moves the margin of adjustment inside the firm: firms produce multiple products and reallocate sales across products when competition changes.
- The paper therefore adds an intra-firm product-composition channel to the trade-productivity literature.

Incremental contribution: The paper's main increment is to show that the unit of adjustment in trade is not just the firm. Exporters respond to market toughness by changing their internal product mix, and this generates new productivity effects.

2.2 Contribution relative to multi-product-firm models

- Earlier models of multi-product firms often emphasize product scope or product cannibalization.
- This paper emphasizes destination-specific competition and its effect on the *relative sales* of products within a firm.
- The model generates variable markups and therefore a direct link from competition to the skewness of a firm's product sales.

- The open-economy environment allows asymmetric countries and geography to matter for competition.

Incremental contribution: The paper adds a tractable multi-country framework in which differences in market size, trade barriers, and supply potential translate into destination-specific product mix choices.

2.3 Contribution relative to empirical exporter studies

- Empirical work had documented that multi-product firms dominate exports and vary product scope across destinations.
- This paper studies a more granular margin: whether the same exporter changes the *distribution* of sales across products depending on destination toughness.
- The paper uses within-firm comparisons across destinations, which reduces concern that the result is driven by persistent differences across firms.
- The empirical results connect directly to theory: tougher competition should raise the sales ratio of better products relative to worse products.

Incremental contribution: The paper provides direct evidence on destination-specific product-mix reallocation using firm-product-destination customs data, rather than inferring the mechanism from aggregate trade flows.

2.4 Contribution for productivity measurement

- Measured firm productivity can change when a firm reallocates labor and sales toward higher-performing products.
- This effect arises even if product-level technologies do not change.
- The paper quantifies the relationship between export-sales skewness and firm productivity using the structure of the model.

Incremental contribution: The paper identifies a new competition-productivity channel: competition changes the composition of production inside the firm, and this can make the same firm appear more productive.

3 Data and measurement

3.1 Firm-product-destination export data

- The empirical analysis uses French customs data on exports by firm, product, and destination.
- A product is defined at a highly disaggregated customs-product level.
- A firm can sell many products to many destinations, allowing within-firm comparisons across markets.
- The baseline empirical analysis focuses on cross-sectional variation across destination markets in 2003.

Interpretation: The data structure is important because it lets the authors compare the same firm selling different product bundles to different destinations. This makes the product-mix margin observable.

3.2 Global and local product rankings

- A firm's **global product rank** is based on its total worldwide sales of each product.
- The firm's **core product** is the product with the highest global sales.
- A product's **local rank** is based on its sales within a destination.
- The authors show that global and local rankings are highly correlated, so the global core product is a meaningful proxy for the firm's best-performing product.

Interpretation: Global rankings avoid mechanical destination-specific selection in defining the core product. Local rankings provide a robustness check and show that the result is not driven by a particular ranking convention.

3.3 Product-mix outcomes

The paper studies two main types of product-mix outcomes.

Top-product sales ratios. For firm f in destination d, the baseline outcome is the log export-sales ratio of the core product $m = 0$ to a worse product m', usually $m' = 1$:

$$y_{fd} = \log\left(\frac{r_{fd}(m=0)}{r_{fd}(m'=1)}\right). \tag{1}$$

Full-distribution skewness. The authors also compute skewness measures using all products exported by firm f to destination d:

- standard deviation of log export sales;
- Herfindahl index of product sales shares;
- Theil index of product sales shares.

Interpretation: The top-product ratio tests the sharpest theoretical prediction. The full-distribution measures show that the result is not confined to the top two products.

3.4 Toughness of competition

The paper uses three main measures of destination competition.

- **Destination GDP:** larger markets attract more entrants and become more competitive.
- **Supply potential:** a market-access measure that captures proximity to large suppliers and therefore the competitive pressure faced in a destination.
- **Freeness of trade:** a bilateral measure recovered from gravity-style trade relationships and used to control for bilateral trade barriers.

Interpretation: Market size alone is not enough because geography matters. A small country surrounded by large suppliers can be very competitive; a large but isolated country may be less competitive.

3.5 Control variables

The empirical regressions include controls for:

- GDP per capita;

- distance;
- contiguity;
- colonial link;
- common language;
- regional trade agreement;
- common currency;
- joint GATT membership;
- bilateral freeness of trade.

Interpretation: The controls separate competition from other features of destination markets that may affect export patterns, such as demand composition, trade barriers, and institutional ties.

4 Theory and main equations

4.1 Preferences and demand

The model uses quadratic preferences over a continuum of differentiated varieties:

$$U = q_0 + \alpha \int_{i\in\Omega} q_i \, di - \frac{\gamma}{2} \int_{i\in\Omega} q_i^2 \, di - \frac{\eta}{2} \left(\int_{i\in\Omega} q_i \, di \right)^2 . \tag{2}$$

This demand system generates linear demand and variable markups. The key feature is that market toughness affects the residual demand faced by each product.

Interpretation: Variable markups are essential. With constant CES markups, tougher competition would not generate the same product-mix skewness mechanism.

4.2 Multi-product technology

Each firm has a core competence indexed by cost c. Products farther from the core have higher costs. A convenient representation is:

$$c_m = \omega^{-m} c, \qquad \omega \in (0,1), \qquad m = 0, 1, 2, \ldots . \tag{3}$$

The core product is $m = 0$. Products with higher m are increasingly far from the firm's core competence and therefore less efficient.

Interpretation: This creates a natural ranking of products within each firm: core products are better, marginal products are worse.

4.3 Operating profits and product cutoff

In a Melitz–Ottaviano environment, a product is produced only if its cost is below the destination-specific cutoff c_D. Operating profits are increasing in the distance between the cutoff and the product's cost:

$$\pi_m(c) \propto (c_D - c_m)^2 \quad \text{if } c_m \leq c_D . \tag{4}$$

Tougher competition corresponds to a lower cutoff c_D.

Interpretation: Lower cutoffs make the market harder to survive in. They reduce markups for all products but disproportionately reduce profits for marginal products.

4.4 Number of varieties produced

The number of varieties produced by a firm is a step function of core productivity:

$$M(c) = \begin{cases} 0, & c > c_D, \\ \max\{m \mid c \leq \omega^m c_D\} + 1, & c \leq c_D. \end{cases} \tag{5}$$

Interpretation: More productive firms produce more products. But tougher competition lowers c_D and therefore causes firms to drop their worst products.

4.5 Competition and product-mix skewness

For two products $m < m'$, the model predicts that tougher destination competition increases the relative sales of product m compared with product m':

$$\frac{\partial}{\partial \text{Competition}_d} \log \left(\frac{r_{fdm}}{r_{fdm'}} \right) > 0. \tag{6}$$

Equivalently, if tougher competition is represented by a lower cutoff c_D, then:

$$\frac{\partial}{\partial c_D} \log \left(\frac{r_{fdm}}{r_{fdm'}} \right) < 0. \tag{7}$$

Interpretation: The product mix becomes more skewed toward the firm's better products in tougher markets.

4.6 Product mix and productivity

Let a firm's measured productivity be a sales-weighted or output-weighted average over products. If tougher competition shifts sales toward better products, then measured productivity rises even if the firm has not improved any product-level technology:

$$Productivity_f = \sum_m s_{fm} \, Productivity_{fm}, \qquad \Delta s_{f,core} > 0 \Rightarrow \Delta Productivity_f > 0. \tag{8}$$

Interpretation: Competition affects firm productivity through internal reallocation across products, not just through selection across firms or process innovation.

5 Empirical specifications and identification

5.1 Baseline product-ratio regression

The main regression relates the log sales ratio of a better product to a worse product to destination-market competition:

$$\log \left(\frac{r_{fd0}}{r_{fdm'}} \right) = \beta_1 \log GDP_d + \beta_2 \log SupplyPotential_d + \beta_3 \log Freeness_{fd} + X_d' \Gamma + \varepsilon_{fd}. \tag{9}$$

The regression is estimated with firm-demeaned data and a random-effects procedure following Wooldridge's approach.

Interpretation: The firm-demeaned design compares the same firm across destinations, so the coefficients capture how destination competition changes the firm's product mix.

5.2 Global-rank and local-rank versions

The paper estimates the ratio using:

- global rankings of products within the firm;
- local rankings of products within the destination;
- alternative comparisons such as core product to the second-best product or the third-best product.

Interpretation: Robustness across global and local rankings shows that the result is not an artifact of how the firm's core product is defined.

5.3 Full-distribution skewness regressions

The authors estimate regressions of the form:

$$Skewness_{fd} = \beta_1 \log GDP_d + \beta_2 \log SupplyPotential_d + \beta_3 \log Freeness_{fd} + X_d'\Gamma + P\left(Products_{fd}\right) + \varepsilon_{fd}, \tag{10}$$

where $P(Products_{fd})$ is a cubic polynomial in the number of products exported by firm f to destination d.

Interpretation: The polynomial controls for mechanical links between skewness measures and the number of products in the exported bundle.

5.4 Why the empirical design is credible

- The unit of observation is firm-destination or firm-product-destination, allowing within-firm comparisons.
- Destination competition is measured using market size and geography, not firm-specific outcomes alone.
- Bilateral trade barriers are controlled using distance, trade agreements, and a constructed freeness-of-trade measure.
- The result is robust to product-ranking definitions and to multiple skewness indices.
- Appendix results replace GDP and supply potential with the number of French exporters in the destination as an alternative competition measure.

Interpretation: The design is not a randomized experiment, but the within-firm cross-destination variation is tightly aligned with the theory's comparative statics.

6 Tables and figures

6.1 Figure 1 and Table 1: product scope and ranking consistency

Read:

- Figure 1 shows that the number of varieties produced is an increasing step function of firm productivity.
- More productive firms produce more varieties because they can profitably sell products farther from their core competence.
- Table 1 shows that global and local product rankings are highly correlated.

- Even when requiring firms to export many products to many destinations, rank correlations remain high, generally above 0.59.

Economic message: The theory's core-product hierarchy is empirically meaningful: firms' best products globally tend also to be among their best products locally.

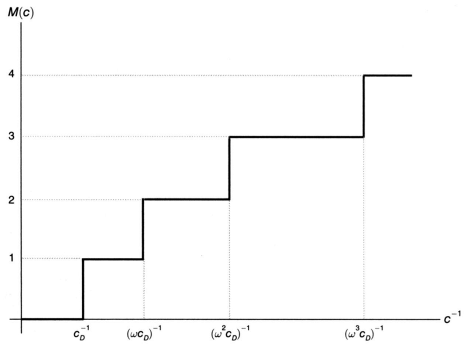

FIGURE 1. NUMBER OF VARIETIES PRODUCED AS A FUNCTION OF FIRM PRODUCTIVITY

TABLE 1—SPEARMAN CORRELATIONS BETWEEN GLOBAL AND LOCAL RANKINGS

Firms exporting at least: to number of countries	Number of products (percent) 1	2	5	10	50
1	67.61	67.47	66.93	65.92	59.39
2	67.58	67.45	66.93	65.93	59.39
5	67.47	67.39	66.93	65.95	59.40
10	67.27	67.22	66.88	65.99	59.46
50	64.48	64.48	64.41	64.12	59.30

Figure 1: varieties produced as a function of productivity *Table 1: global and local rank correlations*

6.2 Figures 2 and 3: destination competition and product-mix ratios

Read:

- Figure 2 shows a positive relationship between mean global ratios and destination GDP.
- Figure 3 shows a positive relationship between mean global ratios and destination supply potential.
- The relationship is visible both for all destination countries and for destinations with many French exporters.
- Larger and more central markets are associated with more skewed sales toward firms' core products.

Economic message: The raw data already show the key comparative static: tougher destinations lead exporters to concentrate sales on their best products.

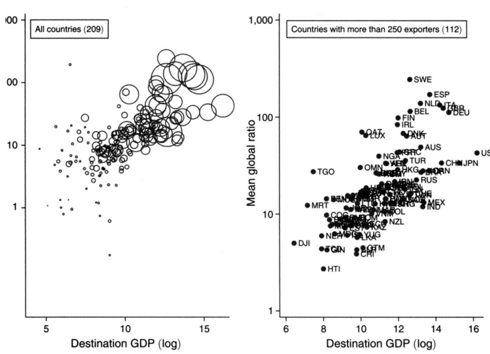

FIGURE 2. MEAN GLOBAL RATIO AND DESTINATION COUNTRY GDP IN 2003

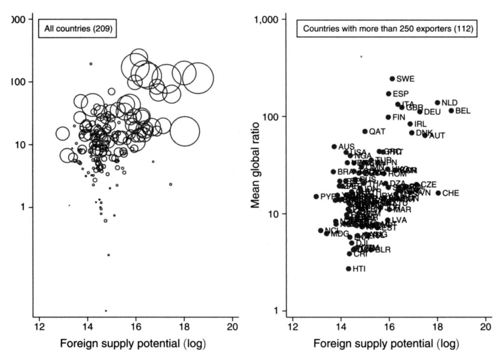

FIGURE 3. MEAN GLOBAL RATIO AND DESTINATION SUPPLY POTENTIAL IN 2003

Figure 2: mean global ratio and destination GDP *Figure 3: mean global ratio and supply potential*

6.3 Table 2: baseline core-to-second-product ratio

Read:

- Table 2 studies the sales ratio of the core product to the second-best product.

- For the global ratio, destination GDP and supply potential are positive and statistically significant in baseline specifications.
- For the local ratio, GDP and supply potential are also positive and significant.
- Bilateral controls and the constructed freeness-of-trade measure do not overturn the main results.

Economic message: The baseline table confirms that firms skew sales toward their core product in larger and more competitive destinations.

TABLE 2—GLOBAL AND LOCAL EXPORT SALES RATIO: CORE ($m = 0$) PRODUCT TO SECOND BEST ($m' = 1$) PRODUCT

	Ratio of core to second product sales' regressions					
	Global ratio			Local ratio		
	(1)	(2)	(3)	(4)	(5)	(6)
n GDP	0.092***	0.083***	0.107***	0.073***	0.057***	0.077***
	(0.013)	(0.012)	(0.010)	(0.008)	(0.005)	(0.006)
n supply potential	0.067***	−0.017	0.044***	0.080***	0.018	0.068 ***
	(0.016)	(0.024)	(0.014)	(0.016)	(0.016)	(0.013)
n distance		−0.063			−0.046*	
		(0.043)			(0.023)	
ontiguity		0.013			−0.108	
		(0.051)			(0.081)	
olonial link		−0.060			−0.041	
		(0.051)			(0.043)	
ommon language		0.023			−0.048	
		(0.050)			(0.038)	
TA		0.066			0.004	
		(0.059)			(0.033)	
ommon currency		0.182***			0.335***	
		(0.047)			(0.036)	
oth in GATT		0.006			−0.033	
		(0.046)			(0.026)	
n freeness of trade			0.096***			0.028
			(0.026)			(0.017)
onstant	−0.000	0.000	−0.000	0.003	0.002	0.002
	(0.016)	(0.012)	(0.014)	(0.012)	(0.011)	(0.013)
bservations	56,097	56,097	56,093	96,891	96,891	96,878
Vithin R^2	0.004	0.006	0.005	0.007	0.011	0.007

Votes: All columns use Wooldridge's (2006) procedure: country-specific random effects on firm-demeaned data, with a robust covariance matrix estimation. Standard errors in parentheses.
*** Significant at the 1 percent level.
** Significant at the 5 percent level.
* Significant at the 10 percent level.

Table 2: core-to-second-product export sales ratios

6.4 Tables 3 and 4: robustness using different product ranks

Read:

- Table 3 uses global rankings and compares the core product with product m'.
- Table 4 uses local rankings and repeats the same exercise.
- GDP and supply potential remain positive and significant across specifications.
- Results hold when the comparison product is the second-best product or a lower-ranked product.

Economic message: The product-mix effect is not confined to the exact core-to-second comparison. The same pattern holds across product-rank definitions.

TABLE 3—GLOBAL EXPORT SALES RATIO: CORE PRODUCT ($m = 0$) TO PRODUCT m'

	(1)	(2)	(3)	(4)	(5)
n GDP	0.107***	0.131***	0.110***	0.096***	0.098***
	(0.010)	(0.010)	(0.011)	(0.012)	(0.011)
n supply potential	0.044***	0.038**	0.038***	0.022*	0.036**
	(0.014)	(0.016)	(0.014)	(0.012)	(0.016)
n freeness of trade	0.096***	0.085**	0.113***	0.137***	0.092***
	(0.026)	(0.033)	(0.032)	(0.038)	(0.026)
n GDP per cap					0.025
					(0.018)
$n' =$	1	2	1	1	1
Destination GDP/cap	all	all	top 50%	top 20%	all
Observations	56,093	22,576	50,623	40,964	56,093
Within R^2	0.005	0.006	0.004	0.002	0.005

Notes: All columns use Wooldridge's (2006) procedure: country-specific random effects on firm-demeaned data, with a robust covariance matrix estimation. Standard errors in parentheses.
*** Significant at the 1 percent level.
** Significant at the 5 percent level.
* Significant at the 10 percent level.

Table 3: global export-sales ratio

TABLE 4—LOCAL EXPORT SALES RATIO: CORE PRODUCT ($m = 0$) TO PRODUCT m'

	(1)	(2)	(3)	(4)	(5)
n GDP	0.077***	0.100***	0.083***	0.061***	0.066***
	(0.006)	(0.012)	(0.011)	(0.016)	(0.008)
n supply potential	0.068***	0.064***	0.051***	0.028*	0.057***
	(0.013)	(0.022)	(0.018)	(0.016)	(0.014)
n freeness of trade	0.028	0.013	0.059	0.092*	0.025
	(0.017)	(0.042)	(0.039)	(0.052)	(0.017)
n GDP per cap					0.029**
					(0.013)
$n' =$	1	2	1	1	1
Destination GDP/cap	all	all	top 50%	top 20%	all
Observations	96,878	49,554	84,708	64,653	96,878
Within R^2	0.007	0.009	0.005	0.002	0.007

Notes: All columns use Wooldridge's (2006) procedure: country-specific random effects on firm-demeaned data, with a robust covariance matrix estimation. Standard errors in parentheses.
*** Significant at the 1 percent level.
** Significant at the 5 percent level.

Table 4: local export-sales ratio

6.5 Table 5: full product-distribution skewness

Read:

- Table 5 uses skewness measures across all products exported by a firm to a destination.
- The standard deviation of log sales, Herfindahl index, and Theil index all increase with destination GDP and supply potential.
- The result holds after controlling flexibly for the number of products exported.
- Freeness of trade is also positively related to skewness in several specifications.

Economic message: The competition effect is not limited to the top two products. Tougher markets reshape the entire product-sales distribution.

TABLE 5—SKEWNESS MEASURES FOR EXPORT SALES OF ALL PRODUCTS

	(1)	(2)	(3)	(4)	(5)	(6)
n GDP	0.141***	0.019***	0.047***	0.052***	0.047***	0.041***
	(0.010)	(0.001)	(0.002)	(0.002)	(0.003)	(0.003)
n supply potential	0.125***	0.016***	0.037***	0.033***	0.023***	0.031***
	(0.023)	(0.002)	(0.004)	(0.004)	(0.004)	(0.004)
n freeness of trade	0.096***	0.007**	0.021**	0.032**	0.045**	0.021**
	(0.036)	(0.004)	(0.009)	(0.013)	(0.022)	(0.009)
n GDP per cap						0.013**
						(0.005)
Dependent variable	s.d. ln x	herf	theil	theil	theil	theil
Destination GDP/cap	all	all	all	top 50%	top 20%	all
Observations	82,090	82,090	82,090	73,029	57,076	82,090
Within R^2	0.107	0.164	0.359	0.356	0.341	0.359

Notes: All columns use Wooldridge's (2006) procedure: country-specific random effects on firm-demeaned data, with a robust covariance matrix estimation. All columns include a cubic polynomial of the number of products exported by the firm to the country (also included in the within R^2). Standard errors in parentheses.
*** Significant at the 1 percent level.
** Significant at the 5 percent level.
* Significant at the 10 percent level.

Table 5: skewness measures for export sales of all products

6.6 Appendix robustness: alternative competition measure

Read:

- Appendix Figure D1 replaces GDP and supply potential with the number of French exporters in a destination.
- Table D1 shows that this alternative competition measure predicts the global sales ratio.
- Tables D2 and D3 show that it also predicts local sales ratios and full-distribution skewness.
- These results support the interpretation that the baseline variables capture toughness of competition.

Economic message: The robustness checks confirm that the results do not depend on one particular competition proxy.

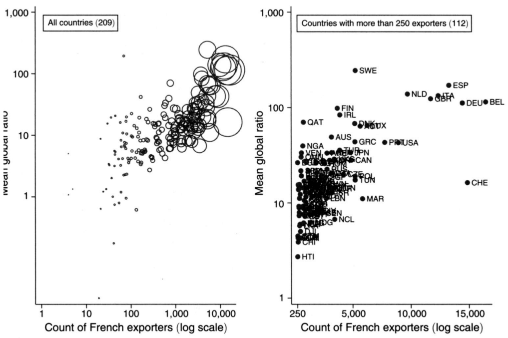

Figure D1: mean global ratio and number of French exporters

TABLE D1—GLOBAL EXPORT SALES RATIO: CORE PRODUCT ($m = 0$) TO PRODUCT m'

	(1)	(2)	(3)	(4)	(5)
n number of French exporters	0.226***	0.263***	0.233***	0.200***	0.200***
	(0.020)	(0.032)	(0.025)	(0.031)	(0.024)
n freeness of trade	−0.034	−0.078***	−0.019	0.018	−0.029
	(0.032)	(0.029)	(0.037)	(0.043)	(0.033)
n GDP per cap					0.031*
					(0.019)
$\imath' =$	1	2	1	1	1
Destination GDP/cap	all	all	top 50%	top 20%	all
Observations	56,093	22,576	50,623	40,964	56,093
Within R^2	0.005	0.005	0.004	0.002	0.005

Notes: All columns use Wooldridge's (2006) procedure: country-specific random effects on firm-demeaned data, with a robust covariance matrix estimation. All columns include a cubic polynomial of the number of products exported by the firm to the country (also included in the within R^2). Standard errors in parentheses.
*** Significant at the 1 percent level.
** Significant at the 5 percent level.

Table D1: global ratio with number of exporters

TABLE D2—LOCAL EXPORT SALES RATIO: CORE PRODUCT ($m = 0$) TO PRODUCT m'

	(1)	(2)	(3)	(4)	(5)
n number French exporters	0.178***	0.210***	0.178***	0.119***	0.129***
	(0.012)	(0.027)	(0.026)	(0.035)	(0.018)
n freeness of trade	−0.056**	−0.096*	−0.026	0.027	−0.040
	(0.026)	(0.050)	(0.045)	(0.058)	(0.026)
n GDP per cap					0.049***
					(0.013)
$\imath' =$	1	2	1	1	1
Destination GDP/cap	all	all	top 50%	top 20%	all
Observations	96,878	49,554	84,708	64,653	96,878
Within R^2	0.007	0.008	0.005	0.001	0.007

Notes: All columns use Wooldridge's (2006) procedure: country-specific random effects on firm-demeaned data, with a robust covariance matrix estimation. Standard errors in parentheses.
*** Significant at the 1 percent level.
** Significant at the 5 percent level.
* Significant at the 10 percent level.

Table D2: local ratio with number of exporters

	(1)	(2)	(3)	(4)	(5)	(6)
n number French exporters	0.348***	0.045***	0.111***	0.118***	0.102***	0.086***
	(0.025)	(0.002)	(0.004)	(0.006)	(0.010)	(0.006)
n freeness of trade	−0.065	−0.014***	−0.034***	−0.027*	−0.007	−0.024*
	(0.048)	(0.005)	(0.012)	(0.016)	(0.020)	(0.012)
n GDP per cap						0.022***
						(0.006)
Dependent variable	s.d. ln x	herf	theil	theil	theil	theil
Destination GDP/cap	all	all	all	top 50%	top 20%	all
Observations	82,090	82,090	82,090	73,029	57,076	82,090
Within R^2	0.106	0.163	0.358	0.356	0.341	0.359

Notes: All columns use Wooldridge's (2006) procedure: country-specific random effects on firm-demeaned data, with a robust covariance matrix estimation. All columns include a cubic polynomial of the number of products exported by the firm to the country (also included in the within R^2). Standard errors in parentheses.
*** Significant at the 1 percent level.
** Significant at the 5 percent level.
* Significant at the 10 percent level.

Table D3: skewness with number of exporters

7 Short summary

- The paper studies how destination competition affects the product mix of multi-product exporters.
- The theory combines multi-product firms with variable markups and asymmetric export destinations.
- More productive firms produce more products, but tougher competition causes firms to drop weak products and sell relatively more of their best products.
- French customs data show that product sales are more skewed toward core products in larger and more competitive destinations.
- The results hold for both global and local product rankings.
- The results hold for core-to-second-product ratios and for full-distribution skewness measures.
- The effects are robust to bilateral controls and to an alternative competition proxy based on the number of French exporters in a destination.
- The model implies that competition-induced changes in product mix can raise measured firm productivity.
- The main contribution is to identify a within-firm product-mix channel linking competition, exporting, and productivity.

8 Conclusion

8.1 Core conclusion

Mayer, Melitz, and Ottaviano show that export-market competition changes not only which firms export, but also how multi-product exporters allocate sales across products. Tougher markets make firms concentrate export sales on their best products and reduce the relative importance of weaker products.

8.2 Economic intuition

A firm's products differ in how close they are to the firm's core competence. In a less competitive destination, even weaker products can earn positive markups and sales. In a tougher destination, weaker products become less profitable, while stronger products remain viable. The firm therefore reallocates sales toward core products. This reallocation changes measured productivity because more resources are used in products where the firm is relatively more efficient.

8.3 Takeaway

The paper broadens the trade-productivity mechanism. Productivity gains from trade are not only about exit of weak firms or entry into exporting. They also arise because multi-product firms change their internal product mix in response to market competition.